1)

$\bar{X} = 52$
$s = 10$
$n = 100$

$SE = \frac{10}{\sqrt{100}} = 1$

$z = \frac{\bar{X} - u}{SE} = 2$

α: 0.1	$1.645 < 2$	reject null
0.05	$1.96 < 2$	reject null
0.01	$2.576 > 2$	fail to reject null

CI:

$\bar{X} \pm 1.96 \cdot 2 = [50.04, 53.96]$

50 outside so we reject.

2)

$\hat{s} = 3.1$
$SE = 1.3$

$z = \frac{3.1 - 0}{1.3} = 2.385$

$\hat{s} \pm 1.3 \cdot \begin{bmatrix} 1.645 \\ 1.96 \\ 2.576 \end{bmatrix} = \begin{matrix} [0.962, 5.239] \\ [0.552, 5.648] \\ [-.249, 6.449] \end{matrix}$

0.1 Reject H_0
0.05 Reject H_0
0.01 Fail to reject H_0

3) $\bar{x} = 11.5$
$s = 3$
$n = 36$
$H_0: \mu = 12$

$SE(\bar{x}) = \frac{3}{\sqrt{36}} = 0.5$

$z = \frac{11.5 - 12}{0.5} = -1$

$\alpha = 0.05$, $11.5 \pm 0.5(1.96)$
$= [10.52, 12.48]$

Fail to reject null if plausible also $|z| < 1.96$.

4) $\bar{A} = 25, s = 8, n = 100$
$\bar{B} = 22, s = 7, n = 100$

$H_0: \mu_A = \mu_B$, $\mu_A - \mu_B = 0$

$SE(A) = \frac{8}{10} = 0.8$

$SE(B) = 0.7$

$\bar{X}_A - \bar{X}_B = 3$ $\quad SE(\bar{X}_A - \bar{X}_B) = \sqrt{\frac{8^2}{100} + \frac{7^2}{100}}$

$1.063 = \sqrt{1.13}$

$z = \frac{3 - 0}{1.063} = 2.822$ $\quad \alpha = 1.96$

Reject!

$3 \pm 1.063 \cdot 1.96 = [0.917, 5.083]$

5) A: $\bar{x} = 14$, $s = 6$, $n = 81$
B: $\bar{x} = 12$, $s = 5$, $n = 64$

$H_0: \mu_A - \mu_B = 0$

$14 - 12 = 2$, $SE = \sqrt{\frac{s^2}{n}_A + \frac{s^2}{n}_B}$

$= \sqrt{\frac{36}{81} + \frac{25}{64}}$

$= 0.914$

$\frac{2-0}{0.914} = 2.189$

$2.189 > 1.96$ Reject H_0!

6) A: $\bar{x} = 3.2$, $s = 1.5$, $n = 49$
B: $\bar{x} = 2.8$, $s = 1.4$, $n = 64$
$H_0: \mu_A - \mu_B = 0$

$3.2 - 2.8 = 0.4$, $SE = \sqrt{\frac{s^2}{n}_A + \frac{s^2}{n}_B}$

$= 0.277$

$z = \frac{0.4 - 0}{0.277} = 1.446$ Fail to reject H_0!

$0.4 \pm 0.277(1.96) = [-0.142, .942]$

7)

a) 93.31%

b) seed = 40, bootstraps = 2000

c) SE = 0.0048

d) $0.9331 \pm 0.0048 \cdot \begin{bmatrix} 1.645 \\ 1.96 \\ 2.576 \end{bmatrix}$

$= [0.9252, 0.9410]$

$[0.9237, 0.9425]$

$[0.9207, 0.9455]$

Reject the null hypothesis at every level, there is not enough evidence to conclude 0.9 is a plausible proportion of homes with central air.

8a)

Estimated Pearson correlation = 0.7232

Seed 40, Bootstraps = 2k

b) SE = 0.01

c) $0.7232 \pm 0.01 \begin{bmatrix} 1.645 \\ 1.96 \\ 2.576 \end{bmatrix}$

$= [0.7067, 0.7397]$

$[0.7035, 0.7428]$

$[0.6973, 0.7490]$

d) typo $p = 0.70$?

if true then

$\alpha = 0.1$	Rejet
0.05	Reject
0.01	Fail to reject

Rejeet means not enough evidence to conclude that null is true while fail to reject means opposite

e) Because they are data points that are toghether.

9a) $\bar{X} = 129.09$ months
seed = 42, bootstrap = 2k

9b) SE = 2.1358 months

9c)

CI	9d)
90% [125.58, 132.61]	Rejeet
95% [124.91, 133.28]	Reject
99% [123.59, 134.60]	Reject

9e) estimates mean record of death or last seen is 129.09 months = 10.76 years all exclude 120 months = 10yrs providing evidence we exceed that value as mean. Doesnt estimate Unrestricted mean lifetime as records end at last follow up.

10a)

Respondents = 5489
$\bar{X}$ = 7.6212 hr
seed = 42, Bootstrap = 2k

10b) SE = 0.0228 hrs

10c/d)

90%: [7.5838, 7.6587] Reject

95%: [7.5767, 7.6658] Reject

99%: [7.5626, 7.6798] Reject

10e) The allowed responders reported average of 7.62 hrs of sleep and all 3 confidence intervals ~~exclude~~ 8 hours so its statistically significant.